

# Shesh Budhabhatti

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## EDUCATION

### NORTHEASTERN UNIVERSITY (Dean's List at Khoury College of Computer Sciences)

Boston, MA

Bachelor of Science in Cybersecurity and Business Administration (Finance)

April 2028

Coursework: Object-Oriented Design, Algorithms & Data Structures, CS Systems, Security Systems, Discrete Structures

## TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, SQL, Assembly

Systems & Tools: Git/GitHub, Linux/Unix, Docker, GDB, Valgrind, Azure DevSecOps, Jira, Bitbucket, Polaron, Miro, Arduino

Security: Wireshark, Metasploit, Hashcat, threat modeling, penetration testing, cryptographic protocols

Spoken: English, Gujarati, Hindi, Spanish

## WORK EXPERIENCE

### MEDTRONIC

Boston, MA

Software Architecture Co-op

August 2026 – Present

- Building an automated pipeline that auto-generates UML diagrams and design docs from code updates for Medtronic's Hugo™ surgical robot platform
- Unifying design docs, diagrams, and database records into one pipeline, migrating documentation to a core database with a cross-team API

Product Security Intern

June 2026 – August 2026

- Built Python tooling to capture and visualize WiFi packet loss across 2.4/5 GHz bands for medical security research
- Analyzed wireless attack surfaces and developed developer security training

### MICROSOFT

Boston, MA

AI Student Ambassador & WICxDevSecOps Program Participant

October 2025 – February 2026

- Selected for security training on Azure DevSecOps pipelines, threat modeling, and AI model security protocols
- Demo Copilot/Omi AI to 100+ users, collecting feedback on quality, privacy risks, and edge cases for product teams

### MEDTRONIC

North Haven, CT

R&D Software Engineering Intern

June 2025 – August 2025 | July 2024 – August 2024

- Reduced test fixture calibration time by 37% through Python automation scripts, saving 4+ hours weekly
- Authored fixture manual for cross-functional teams across US/India (15+ engineers), reducing onboarding time
- Implemented AES-256 and Blowfish encryption protocols for Signia™ Stapling System, ensuring HIPAA compliance and passing FDA cybersecurity audit requirements

### SECURE & ASSURED INTELLIGENT LEARNING LAB

New Haven, CT

Cybersecurity & Safety Researcher

June 2023 – February 2024

- Identified 8 security vulnerabilities in LLM systems through 50+ prompt injection attacks

### KPMG

Hartford, CT

Summer Intern, Audit/Tax/Advisory Accounting

July 2023 – August 2023

- Collaborated with 5-person teams across 3 companies, identifying 2 key control weaknesses in inventory tracking
- Analyzed financial data across 3 client studies with Tableau, visualizing audit risk insight for senior stakeholders

### GURLS CHESS CLUB NON-PROFIT ORGANIZATION

USA

Founder & Executive Director & TEDx Speaker

December 2021 – Present

- Founded chess non-profit coaching 200+ students (70% female); delivered 70+ lessons, TEDx talk/USCF feature
- Redesigned curriculum using retention data analysis, improving participant retention from 40% to 78%

## PROJECTS

### FUSE Filesystem Implementation | C, Systems Programming

November 2025

- Built a custom filesystem supporting large files (>4KB) with inode management and mmap/munmap allocation
- Debugged memory corruption using GDB and Valgrind, achieving 31/31 test cases.

### Memory Allocator & Data Structures | C, Systems Programming

October 2025

- Implemented malloc/free using mmap/munmap with free list management and a first-fit allocation strategy
- Optimized memory reuse through block coalescing, merging adjacent free blocks to minimize fragmentation

### BGP Router Implementation | Python, Networking

February 2026

- Implemented a BGP router handling handshake, update, and withdrawal messages with longest-prefix-match forwarding and a full tie-breaking chain
- Wrote all IP address bit-math from scratch with no external libraries; rebuilt the forwarding table on withdrawal for correctness

## AWARDS/PUBLICATIONS/CERTIFICATIONS

TEDx: "Empowering Equality, Inspiring Action" (2023) | SMARTA Podcast: "If You're Young, You Still Can Make a Difference" (2025) | Lean Six Sigma Yellow Belt (2024) | CT Women's Chess Champ. (2018) | TRAIN's Intro. to Artificial Intelligence Certified (2024)